STANDARD OPERATING PROCEDURE

Inspection of Precision Shaft Assembly - PS-2024-001

SOP No: SOP-INS-2024-031
Rev: B
Effective: 2024-11-01

1. PURPOSE

This procedure defines the inspection steps for Precision Shaft Assembly Part #PS-2024-001.
All critical dimensions must be verified against engineering drawing ED-PS-2024-001 Rev A.

2. EQUIPMENT REQUIRED

- Outside Micrometer (0-25mm range, 0.001mm resolution)
- Bore Gauge (10mm, 0.001mm resolution)
- Height Gauge (0-300mm, 0.01mm resolution)
- Thread Gauge (M12 pitch diameter)
- Profilometer (Ra measurement capability)
- Optical comparator (optional, for thread profile verification)

3. SETUP INSTRUCTIONS

3.1 Verify all measurement instruments have current calibration certificates.
3.2 Clean the workpiece with isopropyl alcohol before inspection.
3.3 Allow the part to stabilize at 20°C ± 2°C for a minimum of 30 minutes.
3.4 Set up the micrometer with zero verification using a calibration standard.
3.5 Prepare the inspection report form (Form QC-INS-001).

4. INSPECTION STEPS

Step 1: Visual Inspection
Examine all surfaces for tool marks, burrs, and surface defects.
Accept/Reject per visual standard VS-001.

Step 2: Outer Diameter Measurement
Measure OD at three positions (25%, 50%, 75% of length) using micrometer.
Specification: Ø25.400 ±0.013mm. Record all three readings.

Step 3: Bore Diameter Measurement
Measure bore ID using bore gauge at two positions.
Specification: Ø10.000 +0.018/-0.000mm. Record readings.

Step 4: Overall Length Verification
Measure overall length using height gauge on surface plate.
Specification: 150.000 ±0.050mm.

Step 5: Thread Pitch Diameter Check
Verify thread pitch diameter using thread gauge.
Specification: M12.700 ±0.025mm. Go/No-Go check required.

Step 6: Surface Finish Verification
Measure surface roughness using profilometer at three locations.
Specification: Ra 0.800 (max 1.600) µm.

Step 7: Final Documentation
Complete inspection report. Verify all dimensions recorded.